Results plan: System One Control of LLM Agent Harnesses with Verifiable Rewards

COMP438/538 project, Halil Kanpak. Working plan of 3 November 2026, not graded. Nothing here has been run.

1 The target

Make Gemma-4-E4B reach the success of Qwen3.8-27B on ALFWorld by putting a trained System One model (Sys1) in front of it. The target is yours. Every other table in this plan supports the middle row below.

Table 1: The project in three rows. Published numbers are from the Bonn benchmark, 134 ALFWorld games, with thinking off and no text generated.

Who plays the game	Model size	Published success (%)	Our success (%)
Gemma alone	8.0B (4.5B effective)	6.7 [3.0, 11.2]	
Gemma + trained Sys1 (the result we want)	8.0B + 4.7B	None	
Qwen 27B alone	27.8B	54.5 [46.3, 62.7]	

Reading key: yellow cell stays empty until measured. gray row is a published number from someone else's setup. A dagger[†] marks a choice Claude proposed, which you can veto. Every choice, who made it, every hyperparameter, the open decisions D1 to D6 and the draft fail conditions are in `choices.pdf` beside this file.

2 Words used in this plan

Table 2: Glossary. These words mean only this throughout the plan.

The players	
Sys1	Picks one option from a list and gives a probability for each option. It writes no text, so it is fast. Same kind of model as Jev. Ours is Kev-4B, trained further
Gemma, Qwen 27B	The two LLMs: Gemma-4-E4B (small) and Qwen3.8-27B (large). An LLM writes text, so it is slow
Harness	The loop around the models. At each step it decides who acts, runs the command and records the output
Gemma + Sys1	The harness with Sys1 in control, handing the steps it cannot take to Gemma
One game	
ALFWorld	A text game of household tasks, such as "put a clean apple in the fridge"
Episode, step	An episode is one game of at most 50 steps. A step is one command
Success	The game's own checker says the task is done. This is the verifiable part of the return
Admissible commands	The commands the game allows at this step, 15 to 44 in ALFWorld
Menu	The short list Sys1 chooses from: the k admissible commands the teacher used most in similar situations, counted on training games. k is a hyperparameter tuned on dev, starting at 10
Fallback	Sys1's menu option that means "LLM, you take this step"
Control state	Three numbers Sys1 sees: who controls now, fallbacks in a row, LLM rounds left
Repeat rate	Share of steps that repeat an earlier command. High means a mistake loop
Handover	One step given to Qwen 27B, one call. Q5 counts these and lasts one step, unlike the hand-over rules

Training	
Teacher	Whoever supplies the right command during training: ALFWorld's built-in expert program, or a strong LLM (decision D1)
Arm	One way of training Sys1. There are five, A0 to A4 (Table 8)
Classification	Train Sys1 to copy the teacher's pick, with no RL
Teacher forcing	During training the teacher's command runs, whatever Sys1 picked
Model in control	During training Sys1's own command runs, mistakes included
Step match	How often Sys1's pick equals the teacher's pick
Confidence	The probability Sys1 gives its own pick
Calibrated	When Sys1 says 80%, it is right about 80% of the time. ECE measures the gap, and lower is better
Exposure gap	Step match on situations the teacher reached, minus step match on situations Sys1 reached by itself
Reading the results	
Dev, test	Dev is 140 games for tuning choices. Test is 134 games, played once at the end
95% interval	The plausible range of a number, given the games played. Written [low, high]
Ablation	Switch one part off, run again and see whether success drops
Bonn, ReDAct	The two main comparison papers. Bonn tests Jev, Gemma and Qwen 27B on ALFWorld. In ReDAct a small LLM hands steps to a large one when unsure, with no training

3 The story, one question per table

Table 3: Each question is asked because the one before it had an answer. G and Q1 check the ground, Q2 and Q3 are the RL part, Q4 is the target, and Q5 to Q7 say what the result costs, protects and depends on.

#	Question	What we measure	Table	Page
G	Do we get the published numbers for Gemma and Qwen 27B in our own setup?	Success on Bonn's 134 games	9	5
Q1	How often is the right next command on the short menu? If often, Sys1 can take those steps and Gemma is left with the hard ones	Share of teacher commands on the menu, and each LLM alone on dev	10	5
Q2	After training, how often does Sys1 pick the teacher's command, and is its confidence honest?	Step match and calibration of each arm	11	6
Q3	Does RL train Sys1 better than plain copying, and should the teacher's or Sys1's commands run during training?	Success of Sys1 alone for each arm, and the exposure gap	12	6
Q4	Does Gemma + Sys1 reach Qwen 27B? (the target)	Success and cost on the test games	13	7
Q5	If a few steps per game may go to Qwen 27B, how much does success rise, and who should choose those steps: chance, Gemma's own uncertainty, or Sys1?	Success at 1, 2, 5 and 10 handovers per game	15	8
Q6	Can Sys1 stop prompt injections by checking each tool call, and does it help that Sys1 never reads the tool output where the attack sits?	Attack success and task success on AgentDojo	16	9
Q7	Which parts of our design actually help?	Ablations of the menu, control state, reward and hand-over rules	17	10

4 What we run

Table 4: Tasks. The harness reads the admissible actions and the success check from the environment on every task. ALFWorld is primary† because Bonn and ReDAct both report on it, its checker is free, and it needs no simulated user. Outside the core: ScienceWorld as an extension (SwiftSage), CyberGym's 10-task subset as counts if a Docker host exists, and BFCL v4 with ToolRet for the reduced project (Clef card, CLM post).

Role	Task and setup	Train / dev / test	Admissible actions	Teacher available for free	Who else reports on it
Primary†	ALFWorld text (`AlfredTWEnv`), 6 task types, 50-step cap	3,553 games / 140 `valid_seen` / 134 `valid_unseen`	Listed by the game, 15 to 44 per step (Bonn)	Built-in expert (`expert_plan`). Its ties use a global random draw, so seed per episode as Bonn does	Bonn on these exact games. ReDAct on 400 episodes
Second, cheap RL†	MiniGrid-DoorKey-8x8-v0, full view as text, 50-step cap	New seeds† / seed 993 / 200 episodes from seed 42	The environment's fixed actions, so menu length does not apply	Shortest-path solver that we write†	ReDAct on this setup. When2Ask on a close variant
Guardrail, and the literal tool calling	AgentDojo, 97 user tasks, 629 security cases, 4 suites	Travel, banking, slack / 20% of those† / workspace†	The suite's tools, plus pass and flag danger	Whether the case carries an injection	AgentDojo's own defences

Table 5: Models. Sizes from the Hugging Face API. Colab L4 has 24 GB. Bonn ran Qwen 27B in bf16 on an 80 GB GPU, and its weights alone come to about 56 GB at that precision, so it does not fit an L4 unquantized. Table 6 has the other combinations.

Role	Model	Size	Licence	Why
Small LLM (your target)	`google/gemma-4-E4B-it`	8.0B, 4.5B effective	Apache-2.0	The model we try to lift. Prior work in Table 19
Large LLM (your target)	`Qwen/Qwen3.8-27B`	27.8B	Apache-2.0	The success we try to reach, and the model behind Q5. Prior work in Table 19
Sys1, primary†	`jaredpalmer/kev-4b` on `Qwen/Qwen3.5-4B-Base`	4.66B base, LoRA adapter	Apache-2.0	No per-option parameters, so the menu can change each step. Documented `-init_from` path for supervised training. The PPO loop is ours
Sys1, ablation	`jaredpalmer/kev-0.8b` on `Qwen/Qwen3.5-0.8B-Base`	0.87B base, LoRA adapter	Apache-2.0	The lightest option, and the only Sys1 that fits an L4 beside Gemma at Bonn's precision (Table 6)
Sys1, ablation	`Mapika/decider-2b`	1.88B	Apache-2.0	Lighter, and its authors report PPO on the served softmax
Sys1, ablation	`Cloudflare/clef-flash`	9.41B	Apache-2.0	Your Cloudflare idea. No official training code, and the third-party fine-tune repo was tested on CPU only
Small LLM, ablation	`Qwen/Qwen3.5-4B`	4.66B	Apache-2.0	Tests whether the result depends on Gemma
Teacher	ALFWorld expert, MiniGrid solver, or a strong LLM	Not applicable	Decision D1	See `choices.pdf`
Reference only	Jev 1.13	Hosted	Closed	Its numbers come from Bonn's repository, so we need no access

Table 6: What fits on an L4 (24 GB). Weights only, Claude's arithmetic from parameter counts, and cache and activations come on top. Gemma measured 14.9 GB in bf16 in arXiv 2604.07035. Every result in this plan states its precision, because quantizing a model can move its numbers away from Bonn's (decision D5).

What runs together	Weights (GB)	Reading
Gemma bf16 alone (G, H1, H2)	16	Fits, same precision as Bonn
Qwen 27B bf16 alone (Bonn's setup)	56	Does not fit. Use Bonn's public responses, or G4
Qwen 27B 4-bit alone (H9, H10, H12)	14	Fits, a different setup from Bonn
Gemma bf16 + Kev-0.8B bf16 (H5 to H7)	17.7	Fits, and keeps Gemma as Bonn ran it
Gemma bf16 + Kev-4B bf16 (H5 to H7)	25.3	Does not fit
Gemma 8-bit + Kev-4B bf16 (H5 to H7)	17.3	Fits, with Gemma quantized
Qwen 27B 4-bit + Kev-4B bf16 (H8)	23.2	Leaves no room for cache
Qwen 27B 4-bit + Kev-0.8B bf16 (H8)	15.7	Fits
Gemma 8-bit + Qwen 27B 4-bit (Q5)	22	Tight. Sequential loading, or G4
Gemma + Sys1 while training A3, A4	Not computed	Adds gradients, optimizer state and a value head. Measured in the first run. A1 and A2 do not need Gemma loaded once its picks are stored

Table 7: One harness step. Your design, with the ALFWorld reading of it.

What Sys1 sees	Task text, first observation, last 5† trace entries including the LLM's last commands and outputs, and the control state
Menu	Every admissible command at this step gets a score: how often the teacher used that kind of command (`go to`, `take`, `heat`) with that object class and place class, in training games of this task type†. The top k form the menu, and k is a hyperparameter tuned on dev. Ranking by kind alone could fill the menu with `go to`, which the expert's search loop leans on (Heuristic, from its code, and Q1 measures it)
What Sys1 can do	Pick a menu command, fall back, abort or flag danger. Abort and flag danger stay off in ALFWorld and MiniGrid until D2 is settled†
What happens next	A pick runs in the game and its output enters the trace. Fallback hands the step to the LLM, which reasons in one call and gives a command in the next, thinking off†, and control returns when it finishes
Hand-over rules	After a set number of fallbacks in a row (start 2) the LLM keeps control for a set number of rounds (start 10). A run of LLM commands that were all on the menu (start 3†) returns control early. All three numbers are hyperparameters
Training labels	The teacher's command at each state, kept only from episodes the checker marks successful†, so "the teacher is right" becomes a checked statement
Training without API calls	Teacher-forced arms replay stored games with no live LLM. Before training we store Gemma's pick at every teacher state, in Bonn's one-pass format†. When the control state gives the LLM a round, the replay scores that pick, not the teacher's, and the control state still moves with Sys1's picks. This is a per-step proxy, because at test Gemma reasons first, sees its own history and keeps control for several rounds. A3 and A4 run Gemma live, which measures the difference
End of episode	Success, abort, or 50 steps

The reward pays Sys1 1 when its pick matches the teacher, and fallback counts as a match when the teacher's command is off the menu. A wrong pick costs w, a needless fallback costs c, and a successful end pays β. Let p be the probability on the top menu command and p_f the probability on fallback. For a fallback lasting one step, with p and p_f taken as true probabilities, picking beats fallback in expected reward when $p(1+w) - w > p_f(1+c) - c$ (Proven, one line of expectation). So w and c set how sure Sys1 must be before it acts, and that rule is only as good as the calibration of p and p_f, which is where calibration enters the return. A fallback that starts several LLM rounds is judged by the return over those rounds, so it has no such closed rule. Under teacher forcing the outcome of the game does not depend on Sys1, so β carries signal only in A3 and A4.

Four rewards are named: R1 is your original agreement reward ($w = c = \beta = 0$), R2 is the default ($w = 1$, $c = 0.1$, $\beta = 0$)†, R3 adds $\beta = 5$† to R2, and R4 pays only $\beta = 1$ at a successful end, which is the purely verifiable return.

Table 8: Training arms. The rule for whose command runs is the research path you named, and A1 to A4 walk along it. RL arms start from A1's weights†, as decider, RelayLLM and ETO start RL from supervised weights.

Arm	Whose command runs	Labels from	Objective	Teacher calls in training	What it tests
A0 untrained	Nothing trained	None	None	0	The starting point
A1 classification	Teacher	Teacher	Cross entropy, then temperature refit on dev	0, stored games	Your classification setting with no RL
A2 RL, teacher forced	Teacher	Teacher	PPO on R2, control state seen	0, stored games	Whether RL adds to A1. With labels known at every state, the step reward gives PPO no signal A1 lacks (Heuristic), so its only addition is the later effect of fallback through the control state
A3 RL, scheduled (DAgger-style mixing)	Teacher, with a probability that falls from 1 to 0†	Teacher at every state reached	PPO on R2, R3 once Sys1 acts	1 per new state, free with the expert. Gemma runs live at fallback	Whether a fading teacher closes the exposure gap
A4 RL, model in control	Sys1	Teacher if free, and the checker	PPO on R3, or R4 with no teacher	1 per state, or 0 under R4. Gemma runs live at fallback	What Sys1's own errors do to learning

Tuning happens on dev and the test games run once after every choice is frozen. Rows play the same games, and intervals come from a bootstrap that resamples games, paired across rows, on each game's mean over seeds. Final arms run 3 seeds. Ablations run 1 seed. With 134 games a paired difference between two policies near 50% success has an

interval of about plus or minus 10 points (Heuristic, assuming they disagree on 40% of games), so differences smaller than that cannot be resolved here. Metrics copy the comparison papers: Bonn's repeat rate, step agreement, ECE and multi-class Brier, and ReDAct's large calls, tokens and steps.

5 The results we aim to fill

G: do we get the published numbers?

Table 9: ALFWorld, 134 games, Bonn's way of playing: the model picks one command from the full admissible list in one forward pass, with the full history in view, thinking off and no text generated. Bonn's responses are public on Zenodo. The first step of each game has the same prompt as Bonn's, so its pick should match even if a later near-tie changes a game. The Qwen 27B row can be checked against those responses if Qwen 27B does not fit our GPU. Draft rule in `choices.pdf`.

Player	Published success	Published steps	Published repeat rate	Our success	Our interval	Our steps	Our repeat rate	First-step picks equal to Bonn's
ALFWorld expert	90.3 [85.1, 95.5]	20.7	0.018					
Qwen 27B	54.5 [46.3, 62.7]	31.6	0.272					
Gemma	6.7 [3.0, 11.2]	47.8	0.541					

Q1: how often is the right next command on the short menu?

Table 10: Top: share of teacher commands that a menu of length k contains, on training games. MiniGrid has no row because its action list is fixed. Bottom: each LLM alone on dev, in the two ways of playing. The stronger thinking-off Qwen 27B row sets the target line[†]. The thinking-on row is a ceiling, not the bar (decision D6). ReDAct does not state thinking on or off, so thinking off in its rows is our pin.

Task and teacher	Commands per game	Kinds of command	$k = 5$	$k = 10$	$k = 20$	All admissible	Fallback rate at chosen k
ALFWorld, built-in expert							
ALFWorld, strong LLM (D1)							
AgentDojo, reference tool calls							

LLM alone, ALFWorld dev	Success	Repeat rate	Tokens per game	Seconds per game
Gemma, picks from the list (Bonn's way)				
Gemma, reasons in one call and commands in the next, thinking off (ReDAct's way)				
Qwen 27B, picks from the list (Bonn's way)				
Qwen 27B, reasons in one call and commands in the next, thinking off (ReDAct's way)				
Qwen 27B, thinking on at its default effort (ceiling)				

Q2: after training, how often does Sys1 pick the teacher's command, and is its confidence honest?

Table 11: Per step, ALFWorld dev, mean of 3 seeds. *Teacher's games* replays the teacher's states, and *own games* are the games Sys1 plays itself. Gray rows are Bonn's untrained models on the test games in their own games, so they belong only in that column. Bonn reports Qwen 27B at 54.5% success with 22.2% step agreement, so on that setup a model can solve games while disagreeing with the expert at most steps. That is why Q2 reports agreement and Q3 reports success.

Arm	Step match, teacher's games	Step match, own games	Falls back when teacher is off menu	Falls back when not needed	ECE	Brier	NLL	Confidence ranks right above wrong (AUROC)
A0 untrained								
A1 classification								
A2 RL, teacher forced								
A3 RL, scheduled								
A4 RL, model in control								
Bonn, Jev 1.13	n/a	0.179	n/a	n/a	0.247	0.987	Not read	Not read
Bonn, Qwen 27B	n/a	0.222	n/a	n/a	0.388	1.084	Not read	Not read
Bonn, Gemma	n/a	0.119	n/a	n/a	0.647	1.410	Not read	Not read

Q3: does RL train Sys1 better than plain copying, and whose commands should run during training?

Table 12: Sys1 alone with no LLM, on the same menu it trained on with fallback masked, dev games, so its success is limited by the menu coverage in Q1. MiniGrid repeats the table.

Arm	Dev success	Minus A1	Match in teacher's situations	Match in its own situations	Exposure gap	Episodes to best dev	GPU hours
A1 classification		Not applicable					
A2 RL, teacher forced							
A3 RL, scheduled							
A4 RL, model in control, R3							
A4 RL, model in control, R4							
ETO, Llama-2-7B-Chat, SFT, test games, ReAct	67.2						
ETO, same model, PPO on final reward, test games, ReAct	29.1						

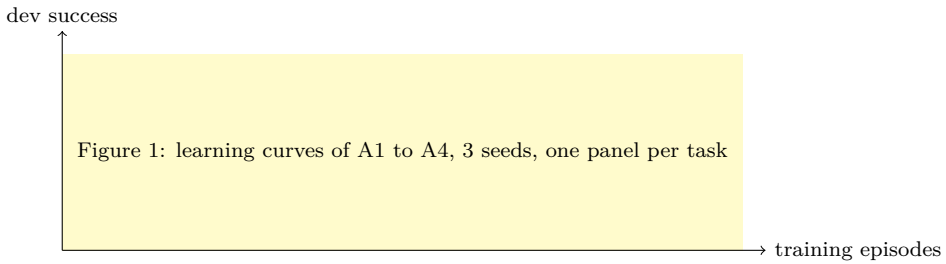


Figure 1: learning curves of A1 to A4, 3 seeds, one panel per task

Q4: does Gemma + Sys1 reach Qwen 27B? (the target)

Table 13: ALFWorld test, 134 games. *Best arm* is chosen on dev in Q3. Calls and tokens are per game. Agent time leaves out the game's own time. *All 3 seeds* is the share of games solved by every seed, which is not τ-bench's pass^k over repeated trials. Gray rows ran other protocols: Bonn is picking from the list, ETO is ReAct with one example on these 134 games, and ReDAct is 400 episodes with no stated split.

Row	Who plays	Success	95% interval	Minus H1	Minus H9	Mean steps	Gemma calls	Qwen 27B calls	Sys1 share of steps	Tokens	Agent time	Repeat rate	All 3 seeds
H1	Gemma alone, picks from the list (Bonn's way)			n/a				0	0				n/a
H2	Gemma alone, reasons then commands, thinking off (ReDAct's way)							0	0				n/a
H3	Sys1 alone, untrained						0	0	1	0			n/a
H4	Sys1 alone, best arm, 3 seeds						0	0	1	0			
H5	Gemma + untrained Sys1							0					n/a
H6	Gemma + Sys1, best of A1 and A2 (teacher forced only), 3 seeds							0					
H7	Gemma + Sys1, best arm, 3 seeds (the result we want)							0					
H8	Qwen 27B + Sys1, best arm, 3 seeds						0						
H9	Qwen 27B alone, picks from the list (Bonn's way)				n/a		0		0				n/a
H10	Qwen 27B alone, reasons then commands, thinking off (ReDAct's way)						0		0				n/a
H11	Gemma fine-tuned to pick the expert's command from the list, all 3,553 games, no Sys1, 3 seeds							0	0				
H12	Qwen 27B alone, thinking on at its default effort (ceiling)						0		0				n/a
	Bonn, Gemma	6.7	3.0, 11.2			47.8						0.541	
	Bonn, Qwen 27B	54.5	46.3, 62.7			31.6						0.272	
	Bonn, Jev 1.13	35.1	27.6, 43.3			37.9			1			0.447	
	ETO, Llama-2-7B-Chat untuned, ReAct with one example	0.0											
	ETO, same model fine-tuned on expert games (SFT)	67.2											
	ReDAct, Qwen3-80B alone	68.3											
	ReDAct, GPT-5.2 alone	78.3											

Table 14: MiniGrid-DoorKey-8x8-v0 test, 200 episodes, ReDAct's seed. Same columns, with the rows that matter most.

Row	Who plays	Success	95% interval	Minus H1	Mean steps	Gemma calls	Qwen 27B calls	Sys1 share of steps	Tokens	Agent time	Repeat rate	All 3 seeds
H2	Gemma alone, reasons then commands			n/a			0	0				n/a
H4	Sys1 alone, best arm, 3 seeds					0	0	1	0			
H7	Gemma + Sys1, best arm, 3 seeds (the result we want)						0					
H8	Qwen 27B + Sys1, best arm, 3 seeds					0						
H10	Qwen 27B alone, reasons then commands					0		0				n/a
	ReDAct, Qwen3-80B alone	85.5										
	ReDAct, GPT-5.2 alone	99.0										

Reading Table 13: H7 against H1 and H2 asks whether Gemma gets better, and H7 against H9 and H10 asks whether it reaches Qwen 27B. H7 against H5 isolates training, H7 against H6 isolates leaving teacher forcing, H4 against H7 shows what Gemma adds to Sys1, and H11 against H7 asks whether Sys1 beats plain fine-tuning of Gemma, the

usual practice in this field. Fine-tuned small models have passed 54.5 on ALFWorld in other formats (ETO 67.2 on these games, AgentGym 77.5 to 83.0 on 200 episodes, AgentTuning 84.0 on 50, each ReAct with thoughts), so Qwen 27B is the bar by your choice and not the limit of the task. H8 against H9 and H10 is your setting where the same large LLM gets shorter and more coherent runs.

Q5: if a few steps per game may go to Qwen 27B, how much does success rise?

Table 15: ALFWorld test success at a fixed number of Qwen 27B calls per game, with thresholds tuned on dev. A handover here is exactly one step, so the hand-over rules are off and Sys1's row sets a threshold on its fallback probability. ReDAct tuned to 5 per episode on calibration episodes, which came to 3.6 on its test episodes. Column 0 reuses H2 and H4.

Who chooses the steps (columns: Qwen 27B calls per game)	0	1	2	5	10	Area under curve
Chance: Gemma plays, random steps go to Qwen 27B						
Gemma's own uncertainty: as in ReDAct, with no training						
Trained Sys1: Sys1 plays, its fallback steps go to Qwen 27B						
ReDAct, Qwen3-80B with GPT-5.2, best uncertainty measure, 400 episodes			80.8 at 3.6 handovers per episode			
ReDAct, Qwen3-80B with GPT-5.2, random steps, 400 episodes			77.0 at 3.7 handovers per episode			

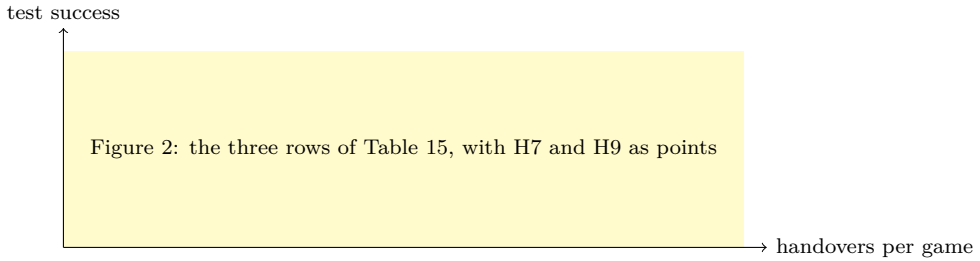


Figure 2: the three rows of Table 15, with H7 and H9 as points

Q6: can Sys1 stop prompt injections by checking each tool call?

Table 16: AgentDojo workspace suite, attack `important_instructions` (the paper's main attack), with `tool_knowledge` as a second run, Gemma as the agent. A *prompt injection* is a hidden instruction in a tool's output that tries to redirect the agent. Sys1 sees each tool call before it runs and answers pass or flag danger, and a flagged call does not run. S5 tests your claim that independence from the LLM's context helps, because an injection in a tool output cannot reach a Sys1 that never reads tool outputs. The reward comes from the suite's own security and task checks.

Row	Defence	Task success, no attack	Task success, under attack	Attack success	False flags per 100 calls	Added seconds per call
S1	No defence				Not applicable	0
S2	AgentDojo `tool_filter`				Not applicable	
S3	AgentDojo `PromptInjectionDetector`					
S4	Sys1 reading the full context					
S5	Sys1 reading only the task and the proposed call					
	Ctrl-Z, BashBench, a different threat model			58 to 7	5% benign cost	
	Task Shield, AgentDojo, GPT-4o, checks each call against the user's goal			Attack success 2.07, task utility 69.79		
	CaMeL, AgentDojo, control flow fixed from the trusted query			77% of tasks solved with provable security, 84% undefended		

Optional, CyberGym 10-task subset, level 1 (counts out of 10, only with a Docker host)	Gemma alone	Gemma + Sys1	Qwen 27B alone
Tasks solved			

Q7: which parts of our design actually help?

Table 17: ALFWorld test, 134 games, Gemma + Sys1, one part changed at a time, 1 seed, run after every choice is frozen, and no choice is made from this table. One seed on 134 games cannot resolve differences under about 12 points (Heuristic), so read this table as direction only. The papers we read hand one step or one move to the large model and none keeps it in control for several rounds, so the hand-over row tests what is distinctive in your design. Tier 1 is required. Tier 2 runs if time remains. R3 and R4 need Sys1 to act, so those rows run on A4.

The part being tested	Default	Changed to	Tier	Success	Gemma calls	ECE	Repeat rate
Full system	All defaults	None	1				
Control state	Seen by Sys1	Removed	1				
Menu length	k tuned on dev, start 10	$k = 5$	1				
		$k = 20$	1				
		All admissible commands	1				
Reward	R2	R1, your agreement reward	1				
		R3, adds success	1				
		R4, success only	1				
Calibration step	Temperature refit	None	1				
		Brier term in the objective	1				
Teacher	Built-in expert[†]	Qwen 27B games	1				
Hand-over rules	2 fallbacks, 10 rounds	None, fallback lasts one step	1				
Hand-over rules	2 fallbacks, 10 rounds	1 fallback, 10 rounds	2				
		2 fallbacks, 5 rounds	2				
Trace window	Last 5	Last 1	2				
		Last 20	2				
		Full history (Bonn)	2				
LLM's last trace in Sys1's view	Seen	Removed	2				
How fallback is chosen	Menu option	Threshold on confidence	2				
		Option and threshold	2				
Teacher	Built-in expert[†]	Birch games, only if collected under D1	2				
Sys1 model	Kev-4B	Kev-0.8B	2				
		decider-2b	2				
		Clef-flash	2				
Small LLM	Gemma	Qwen3.5-4B	2				

6 Budget and order of work

Table 18: Runs and games. Gemma and Qwen 27B are open weights, so only a Birch teacher (D1) uses paid calls. GPU hours are filled from the first run, which goes end to end before any sweep. One 134-game pass in Bonn's way took 4,236 requests at 1.48 s for Qwen 27B on an A100 and 6,410 at 0.27 s for Gemma on an A40, and rows that generate text cost more. Colab: you stated about 600 hours a month. Third-party pages say Pro+ gives 500 to 600 compute units and an L4 burns 1.71 per hour, which is about 290 to 350 hours (Open: check your Colab page).

Experiment	Size	Training runs	Games played	GPU hours
Teacher games (D1)	Expert: all 3,553 games, free. A paid LLM: 500† games, at most 25,000 calls	0	3,553 or 500	
Gemma's stored picks	One forward pass at about 73,500 expert states (3,553 games, 20.7 steps), about 5.5 hours at Bonn's 0.27 s on an A40 (Heuristic). A subset if an L4 is slower	0	0	
G	3 players, 134 games	0	402	
Q1	5 LLM rows, 140 dev games	0	700	
Q2, Q3: A1, A2	2 arm settings, 3 seeds, 2 tasks, no live LLM	12	Dev only	
Q2, Q3: A3, A4	3 arm settings (A4 under R3 and R4), 3 seeds, 2 tasks, Gemma live	18	Dev only	
Q4 ALFWorld	12 rows, 20 runs with seeds, 134 games. H1 and H9 reuse G	3	2,680	
Q4 MiniGrid	5 rows, 11 runs with seeds, 200 episodes	0	2,200	
Q5	3 rows, 1, 2, 5 and 10 handovers, 134 games	0	1,608	
Q6	5 rows, S4 and S5 at 3 seeds, workspace cases	6	Case counts	
Q7	11 changes in Tier 1 and 13 in Tier 2, 134 games each. The small-LLM row and the threshold rows need no retraining	21	3,216	

The order follows the story: G, Q1, one run end to end, Q2 and Q3 on ALFWorld, Q4 on ALFWorld, then Q5, Q6, MiniGrid and Q7. The syllabus and guidelines give no dates for the later milestones, so this mapping is by content†. The Formulation and Design Report carries the harness, the arms and G. The Progress and Evaluation Report carries Q1 to Q4 on ALFWorld. The presentation and final report carry the rest.

The cut order when time runs short† is ScienceWorld, CyberGym, MiniGrid, Tier 2, Q6, A3. G, Q1, the arms A1, A2 and A4, and Q4 on ALFWorld survive every cut. The last resort is your reduced project: a System One tool selector on BFCL v4 with A0, A1 and A2, which still measures the effect of RL training.

Appendix: changes from the proposal

The proposal says the formulation can change, and the Formulation and Design Report is where the change gets stated. Tasks move from τ-bench to ALFWorld and MiniGrid, so the simulated user leaves the environment. A menu of frequent tool calls becomes frequent kinds of command filled in from the admissible commands, which settles the open question on arguments. The small LLM is now Gemma and the strong second setting is Qwen 27B. CyberGym moves to optional and AgentDojo carries security, as you suggested with "another task for guardrail part". Sys1's action set, the hand-over rules, the fallback label, teacher forcing as a research path, the control state and the comparison with the LLM alone are unchanged. ALFWorld commands are not tool calls, so Q6 is the only literal tool-calling evidence in the core plan (Open whether Q4 transfers to it). The course requires an LLM-use disclosure with each deliverable, and this plan should appear in the next one.

Appendix: comparison papers

Table 19: Why these two models: who else uses them. Only Bonn uses both on one task, and only Bonn uses either on ALFWorld. When2Ask, SwiftSage and ReDAct use neither. We found no work that trains a Sys1 to lift Gemma. This comes from web searches, so a missed paper is possible. Bellman Language Models and MobilePA-Bench also use Qwen 27B and are in the next table. All numbers are Empirical on the setup stated.

Work	Uses	Task, and how the model runs	Published number
Bonn, arXiv 2609.37647	Both	ALFWorld and 37 datasets, as reference models beside Jev. Thinking off, one forward pass	ALFWorld success: Gemma 6.7, Qwen 27B 54.5
Gemma model card	Gemma	τ^2, settings not given	42.2% average over 3
Small coding agents, arXiv 2610.33958	Gemma	SWE-bench Verified issue localization, tool ablation, models from 0.6B to 32B	Function Acc@10 on 500 tasks: 16.7 with grep tools, 29.7 with all tools
Not Every Call Needs a Frontier Model, arXiv 2610.09021	Gemma	Five call sites of a home-automation agent, one of seven local models, 4-bit	Overall 78.6, best local model for one call site
Dense and MoE study, arXiv 2604.07035	Gemma	ARC, GSM8K, Math, TruthfulQA. No agent	Mean VRAM 14.9 GB
Clef, Kev, JEV-27B	Qwen 27B	Open Sys1 models built on it or offered on it. Your Cloudflare idea starts here	Clef card numbers in the appendix

Table 20: All numbers are Empirical, on the setup stated, and none predicts ours. The last column is what we copy so our rows read beside theirs.

Work	Tasks	Method and models	Published result	What we copy
Bonn/Lamarr, arXiv 2609.37647	ALFWorld unseen, 134 games, 50 steps. Also TextWorld, BabyAI-Text, FrozenLake	Untrained pick over admissible commands. Jev 1.13, Qwen 27B, Gemma. MIT code and raw data	Success: Jev 35.1 [27.6, 43.3], Qwen 27B 54.5 [46.3, 62.7], Gemma 6.7 [3.0, 11.2], built-in expert 90.3	Games, state, success, repeat rate, step agreement, ECE, Brier, and G
ReDAct, arXiv 2604.07036	ALFWorld 400 episodes, 50 steps. MiniGrid DoorKey-8x8, 200 episodes	Threshold on action-level uncertainty, no training, tuned to about 5 large calls per episode. Small Qwen3-80B and others, large GPT-5.2 and others. No code link	ALFWorld: Qwen3-80B 68.3, GPT-5.2 78.3 with 9,898 large calls, best handover 80.8 with 1,447, random handover 77.0 with 1,487. MiniGrid: Qwen3-80B 85.5, with handover 93.0 to 95.0, random handover 93.0, GPT-5.2 99.0	Reasoning then a command, the uncertainty and random rows of Q5, MiniGrid seeds, cost columns. Its plus-minus values are bootstrap standard deviations, and its 400 ALFWorld episodes have no stated split
ETO, arXiv 2403.02502	ALFWorld 134 unseen games, also WebShop and ScienceWorld	Llama-2-7B-Chat fine-tuned on 3,321 expert instances, then contrastive exploration. ReAct with one example	Untuned 0.0, SFT 67.2, PPO 29.1, ETO 72.4, GPT-4 38.1	The fine-tuning row H11, and evidence that imitation is the usual start
AgentTuning, arXiv 2310.12823	AgentBench's ALFWorld, 50 instances	Llama-2 fine-tuned on 336 success-filtered GPT-4 ReAct trajectories out of 954	7B chat 2.0, AgentLM-7B 84.0, GPT-4 78.0	Filtering teacher games by success (our C8)
AgentGym, arXiv 2406.04151	ALFWorld, 200 evaluation instructions	Behaviour cloning on 2,420 trajectories, then agent evolution	Llama-2-7B 2.0, BC 77.5 and 83.0, GPT-4-Turbo 67.5	Same
BUTLER, ALFWorld paper, arXiv 2010.03768	ALFWorld, 50K episodes	DAgger with the handcoded expert, whose share falls from 100% to 1%, 8 seeds, text generation	Read at setup	The schedule idea in A3
TWOSOME, arXiv 2401.14151	VirtualHome and Overcooked, up to 500k steps	PPO with LoRA on frozen LLaMA-7B, scoring all valid actions by likelihood, 3 seeds	Reports better sample efficiency than plain PPO and SayCan	The closest method to A4
GLAM, arXiv 2302.02662	BabyAI-Text, 1.5M steps, 6 actions	PPO on Flan-T5 780M scoring valid actions, 4 seeds	Read at setup	RL budgets on action lists
Bellman Language Models, arXiv 2610.24871	τ^2 telecom, SymPy issues, Mozilla CI triage, InventoryBench	Controller built from a solved decision process over discretized conversation states, from logs or the agent's own runs. Moves fixed by the state are sent directly, and unseen states go back to the LLM. The abstract names no RL. Preregistered, no code link	Telecom, tasks solved of 114, Qwen 27B alone then with BLM: 52 to 94, 31 to 102, 33 to 109. Same set as 31 to 102, Qwen 27B alone with thinking 62. SymPy 14 to 19 of 63, not significant	Closest recent work on a controller around an LLM. No Gemma, no ALFWorld
MobilePA-Bench, arXiv 2608.23035	Mobile planning sandbox	Qwen 27B trained with online GRPO, 4,000 queries, one exploratory checkpoint	Overall score 66.94 to 69.25	Precedent for RL on Qwen 27B itself
When2Ask, arXiv 2306.03604	MiniGrid variants, Habitat	PPO policy on when to ask, penalty when the plan does not change. Vicuna-7b, 13b. Baselines Hard-coded, Always, Random, Never	Reports LLM interaction counts	RL on when to call the LLM
SwiftSage, arXiv 2305.17390	ScienceWorld, 30 task types	T5-large by behaviour cloning, generative, with GPT-4 and a heuristic switch	Average score 84.7	Extension task and its score
RelayLLM, arXiv 2601.05167	Math benchmarks	Qwen3-0.6B, 1.7B call Qwen3-8B. Supervised warm-up, then GRPO, reward is correctness minus call ratio	Qwen3-1.7B 42.50 to 49.52 with 1.07% of tokens, teacher alone 54.12	Reporting token share beside accuracy

Table 21: Comparison papers, continued: System One vendors and security. Same reading as the table before.

Work	Tasks	Method and models	Published result	What we copy
decider v8 to v10	16 live MiniWoB++ click tasks and games	PPO on the served softmax for 384 steps after supervised training, with a KL term to the supervised weights. Recipe in `docs/RL.md`, training code kept separate	Vendor report	PPO on a softmax over options
Clef card, CLM post	BFCL, ToolRet, BANKING77. BFCL v4, T-Rex, WikiRacing, Mario	Schema head, or state-action scorer on Qwen3-8B. Supervised, as Kev's card also describes, with a fitted temperature. Self-reported	Clef: BFCL 98.5, ToolRet 69.2. CLM: "on par" with Jev	Tasks for the reduced project
AgentDojo, arXiv 2406.13352	97 user tasks, 629 security cases	`tool_filter`, `PromptInjectionDetector`, main attack `important_instructions`	Read from its results at setup	The suite, the checker, the defence rows
Ctrl-Z, arXiv 2504.10374	BashBench, 257 tasks	Control protocols that resample suspicious commands	Attack success 58% to 7% at 5% benign cost	Reporting attack success beside benign cost
CyberGym, ICLR 2026	1,507 tasks	Measures exploitation capability	Read at setup if run	Counts on its 10-task subset